Empirical Proof Record: VALIDATED

2026-05-23T23:11:38.757051+00:00

Empirical Proof Record — VALIDATED

Proof ID: `31deaa26470a5378e89994f0ba757714…`
Created: 2026-05-23T23:11:38.757051+00:00

1. Claim

Across 30 synthetic (x, y) pairs of size 100 (seed=20260518), "scipy.stats.pearsonr", "numpy.corrcoef", and "pingouin.corr(method='pearson')" produce Pearson r values that agree to ≤ 1e-09 across all three pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer; three-way variant.)

- **Operationalisation:** Generate N (x, y) pairs where x ~ Normal(0, 1) and y = ρ_target × x + sqrt(1 - ρ_target^2) × Normal(0, 1) for a sweep of ρ_target across [-0.9, 0.9]. Compute Pearson r under each of scipy, numpy, pingouin. Compute max_abs_diff = max over (pair, backend-comparison) of |r_a - r_b|. VALIDATED iff max_abs_diff ≤ tolerance across all three pairwise comparisons.
- **Threshold:** `max_absolute_pearson_difference <= 1e-09 correlation`
- **H0:** At least one backend pair (scipy↔numpy, scipy↔pingouin, numpy↔pingouin) disagrees on Pearson r by more than the floating-point tolerance.
- **H1:** All three backends compute identical Pearson r to floating-point tolerance across every pair.

2. Verdict

VALIDATED — observed `3.3306690738754696e-16`
30 pairs × 3 backends; max pairwise |Δr| = 3.331e-16 on scipy_vs_numpy (≤ 1e-09 tol)

3. Evidence

Pillar	Statistic	Valu
pearson_scipy_vs_numpy_vs_pingouin	max_absolute_pearson_difference	3.3306690738754696e-1

4. Pre-registration

- Registered at: 2026-05-23T23:11:38.544957+00:00
- Config hash: 395ced3ae94d681473ffeddcaa298fd2…
- Data hash: 395ced3ae94d681473ffeddcaa298fd2…

5. Signature

65db23e787b18f0046d75edb36b2e52664ae764de7f5c84c4a1fdbce8c7b99e3